The corner store gap: SNAP food stores the map cannot see

Oct 30, 2026 · @Asfandyar Farooqi

In the most socially vulnerable quarter of census tracts, one in ten stores that accept SNAP food benefits is missing from Overture. In the least vulnerable quarter it is one in seventeen. Chain stores are found equally well everywhere: the gap is the independent stores those neighbourhoods rely on, and the coverage gap score cannot see it.

Entry

Item	Detail
Zindi user	AsfandyarFarooqi
Prizes entered	Best Bias Discovery and Best Documentation
Scored submissions (selected for the private board)	6AMWPqPS, blend A, public RMSE 0.073886708; tDuujvN9, blend B, public RMSE 0.074494132
Code	github.com/AsfandyarFarooqi/bias-bounty-lifelines
Where each prize's material is	Best Bias Discovery: from "the finding" to Limitations. Best Documentation: from "how our leaderboard estimate is built" to the Compliance statement. Reproducibility and data sources serve both.

Best Bias Discovery: the finding

USDA publishes every store authorized to accept SNAP, the federal food benefit for low-income households, with its name, address, store type and coordinates. We matched each SNAP storefront in the five study regions against the Overture places in the challenge data (release 2026-08-19.0). A store counts as on the map when an Overture place nearby carries its name or sits at its street address.

- **2,318 of 25,221 SNAP storefronts (9.2%) have no matching place in Overture.**

- **The gap rises with social vulnerability.** By the tract's CDC Social Vulnerability Index, 5.9% of stores are missing in the least vulnerable quarter, then 7.7%, 9.0% and 10.5% in the most vulnerable quarter: 1.8 times as many (95% interval 1.5 to 2.1).
- **Chain stores are the control, and they stay flat:** 4.0% to 5.2% missing in every quarter. The method finds a chain store equally well in any neighbourhood.
- **Independent stores carry the gap.** They are missing 15.3% of the time, three times the chain rate, rising from 11.2% to 16.4% with vulnerability. They are also a quarter of SNAP stores in the least vulnerable tracts (26%) and half in the most vulnerable (49%).
- **People live with it.** 1,926 tracts, home to 8.6 million people, have at least one SNAP store missing. In 190 tracts (727,000 people) every SNAP store that USDA lists is missing.

The stores counted are those authorized by 30 June 2025, over a year before the Overture release, and still authorized in October 2026. Farmers markets, delivery routes and wholesalers are left out.

Why the scorecard cannot see it

The coverage gap score asks how many places a tract has, never which ones.

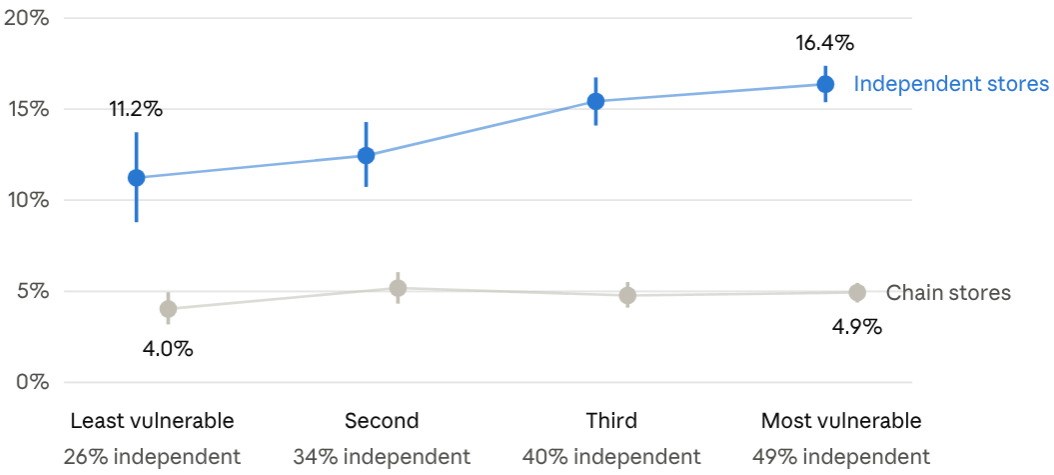
- **Business places are a head count.** The establishments half divides the number of Overture places in a tract by the number of employer establishments in County Business Patterns, and caps the ratio at 1. Tracts with a missing SNAP store have as many Overture places per resident as tracts without one (median 42 against 40 per 1,000 residents), so the ratio cannot tell them apart.
- **The reference leaves out owner-run shops.** County Business Patterns covers only "establishments with paid employees" (Census), while "the majority of all business establishments in the United States are nonemployers" (Census). A family store with no paid staff is not in the reference at all, so its absence from Overture cannot register.
- **The facilities half looks at three types only:** fire stations, EMS stations and schools. Food stores are not among them.

The Bias Scorecard compares these scores across strata, so it inherits the blind spot. A tract can lose every SNAP store from the map and still look fully covered.

Evidence

Independent SNAP stores go missing more often as vulnerability rises; chains do not

Share of SNAP storefronts with no matching place in Overture, by the tract's CDC Social Vulnerability Index quarter. Dot: estimate. Vertical line: 95% interval (bootstrap over tracts).



USDA SNAP retailers retrieved 30 October 2026, matched to Overture places 2026-08-19.0; 25,221 storefronts in 7,935 tracts

Independent stores (blue) are missed 11.2% of the time in the least vulnerable quarter and 16.4% in the most vulnerable. Chain stores (grey) stay between 4.0% and 5.2%. Underneath, the store mix shifts: independents grow from 26% to 49% of SNAP stores.

It holds however a match is defined

Matching rule	Counts as on the map	Missing, least vulnerable quarter	Missing, most vulnerable quarter	Chains, least to most
Strict	Same name within 150 m (score 90 or more), or same street address and name	13.8%	22.1%	9.1% to 10.5%
Main, used above	Same name within 150 m (80+) or 500 m (90+), a distinctive name within 3 km (95+), or same street address	5.9%	10.5%	4.0% to 4.9%

Matching rule	Counts as on the map	Missing, least vulnerable quarter	Missing, most vulnerable quarter	Chains, least to most
Lenient	Main, plus any food store, fuel station or pharmacy within 50 m under another name	3.1%	5.9%	2.7% to 3.3%

Even under the lenient rule, which counts any food place at the spot as the store, independent stores in the most vulnerable quarter are missing twice as often as in the least vulnerable (8.8% against 4.1%).

It holds inside each region and in cities alone

Region	Independent stores missing, least vulnerable quarter	Most vulnerable quarter
maricopa-az	8.3% of 60	18.2% of 505
northern-ca	4.6% of 65	12.7% of 574
south-central-tx	12.3% of 454	17.6% of 3,839
eastern-ok	14.8% of 61	15.1% of 516
eastern-wa	8.1% of 37	9.6% of 417

In urban tracts alone (19,804 stores), chains run from 3.8% to 4.8% and independents from 11.3% to 17.0%. Among stores authorized before mid-2021 (18,136), chains run from 2.8% to 3.5% and independents from 10.3% to 12.4%, so new openings do not explain it.

Controls

A logistic regression on the 10,512 independent stores controls for store type, region, rural status, tribal land, distance to a HUD colonia and population density, with errors clustered by county. Going from the least to the most vulnerable tract multiplies the odds of being missing by 1.67 ($p = 0.008$). Rural location slightly lowers the odds (0.78); tribal land and colonia proximity show no effect. Across all 25,202 stores with complete data, a chain store has 0.42 times the odds of being missing ($p < 0.001$).

Checked by hand

We read the Overture places around 40 random misses: 20 in the most vulnerable quarter and 20 in the two least vulnerable. 23 have no plausible counterpart, 10 are on the map under a fuel brand or another name or far from USDA's point, and 7 are unclear. False misses ran at 6 of 20 in the most vulnerable quarter and 4 of 20 in the least vulnerable half; two of the six were tribal-land stores whose USDA point is far from the store.

We then looked up 20 of the missing stores on Google Maps, 14 in the most vulnerable quarter and 6 in the least vulnerable half. All 20 appear at the address USDA lists, 19 of them with posted opening hours. They are real, trading stores that Overture does not show.

Six of the missing stores

Store, as USDA lists it	Place	Tract SVI	What Overture shows at the spot	Check
77mart	Fort Worth, TX	0.99	A title-loan office at 39 m, a truck rental at 64 m	Google Maps, Overture
La Familia Padron	Brownsville, TX	0.98	Nothing within 100 m	Google Maps, Overture
Carniceria Los Reyes 3	Phoenix, AZ	0.98	A Dollar General at 20 m, a money transfer office at 25 m	Google Maps, Overture
Midway Food Mart	Tyler, TX	0.92	A historic house at 53 m, an insurance agency at 56 m	Google Maps, Overture
La Cubana Latin Market And Bakery	Arlington, TX	0.88	A chiropractor at 19 m, a barber at 23 m, two childcare centres	Google Maps, Overture
Fresh Latin Market	Big Spring, TX	0.86	A supplement shop at 5 m, an ice cream shop at 30 m	Google Maps, Overture

Who is affected and why it matters

SNAP is how low-income households buy food, and in the tracts where the gap is widest they rely on independent stores: half of all SNAP stores there are independent. A family without a car shops at the store within walking distance. If that corner store or carnicería is missing from Overture, it is also missing from the apps, delivery services and planning

tools that rely on Overture's places. Disasters make this sharper: SNAP rules let households replace food lost to fire, flood or long power cuts (MassLegalServices), but that help only works at a store people can find. I spent three years at Pakistan's National Disaster Management Authority, and in any relief operation one of the first questions is where people can still get food. A gap of a few percent looks small in a national average, but it falls on the neighbourhoods with the fewest other stores, where finding food matters most.

Recommendations

For Overture

1. Conflate the USDA SNAP retailer list into the places theme. It is federal public data with names, street addresses, store types and coordinates, and USDA keeps it current. Independent food stores are where Overture's other sources are thinnest.
2. Report place completeness by category against public registries in each release, split by social vulnerability, so a gap like this shows before users meet it.
3. Where a fuel brand sits over a store with its own name, keep both names. 10 of our 40 hand-checked misses were stores listed under a brand, another name or a distant point.

For the scorecard

1. Score places by type against type-specific registries, for example food stores against the SNAP list, instead of one total count against employer establishments.
2. Compute the establishments half per category, or drop the cap at 1, so a surplus of other places cannot hide missing grocers.
3. Add the share of SNAP stores on the map to the Bias Score, reported across its existing strata.

For anyone planning food access after a disaster

1. Start from USDA's own retailer list, not from general map data, when telling residents where benefits can be spent or deciding where to send food.

Reproducibility

Every number in this write-up comes from two Colab notebooks that read the public challenge bucket directly, plus scripts that run on their outputs. The only randomness is a seeded bootstrap.

1. **bias_bounty_features_v2.ipynb** (about 25 minutes): per-tract Overture measurements for all five regions and the strata tables. This feeds the scored file.

2. **bias_bounty_lifelines.ipynb** (about 5 minutes): downloads the USDA SNAP retailer list, current and historical, and matches every store against Overture places by name, street address and distance. It writes every score and distance, the Overture places around a sample of stores, and a sources file with each download's URL, retrieval time and SHA-256.
3. **Analysis scripts:** analyze.py (rates, bootstrap intervals, regressions), audit_review.py (the hand-check sheet) and blend_run.py (the leaderboard calibration).

Code, notebooks, results and this write-up as a PDF: github.com/AsfandyarFarooqi/bias-bounty-lifelines (MIT licence).

Data sources

Source	Used for	Retrieved
Challenge bucket on Source Cooperative: Overture 2026-08-19.0 extracts, census tracts, ACS B25001, strata tables	The scored file, and the Overture side of the finding	29 October 2026
USDA SNAP retailer locations, current, read from USDA's ArcGIS service	Bias Discovery: the store list, 33,900 records in the study regions	30 October 2026, 09:19 UTC
USDA Historical SNAP Retailer Locator Data, 2005 to 2025	Bias Discovery: authorization dates and detailed store types	30 October 2026, 09:19 UTC
HUD Colonias Communities	Bias Discovery: distance from each store to the nearest colonia, a regression control	29 October 2026
HRSA Health Center Service Delivery and Look-Alike Sites	Tested, not used for a finding (see Limitations)	30 October 2026, 09:19 UTC

Limitations

- **The registry has its own errors.** USDA geocodes each store's address, and on tribal land and rural routes the point can sit kilometres away. The Bashas' supermarket in Sells, on the Tohono O'odham Nation, is placed 12 km from the store that Overture does show. Chain stores are our check on this: their miss rate does not change with vulnerability.

- **Stores open and close.** We keep only stores authorized by June 2025 and still authorized in October 2026, so a closure would have to fall inside that window. Stores authorized before mid-2021 show the same pattern.
- **Missing means no Overture place with the store's name nearby or at its street address.** Some stores trade under a fuel brand or a new name. The lenient rule bounds that case, and the gap holds under it.
- **Most stores sit in vulnerable tracts.** The quarters follow the national SVI ranking, so the least vulnerable quarter is the smallest group (2,639 stores against 12,033 in the most vulnerable).
- **This is an association across tracts, not a cause.** We do not know why Overture's sources miss these stores; a thin online presence for small shops is a likely reason, not a tested one.
- **One registry did not work.** We also matched 1,494 HRSA community health center sites. That list mixes clinics with offices, mobile units and classrooms in other buildings, so we did not use it for a finding.

Best Documentation: how our leaderboard estimate is built

Our scored file estimates each tract's coverage gap from permitted data only. The Overture side of every ratio is measured exactly, each withdrawn reference count is replaced by its expected value, and the pieces are combined by the official rule: the mean of the defined components.

Data

Everything is read straight from the challenge bucket on Source Cooperative with anonymous S3 access, in batches, inside Google Colab: the Overture 2026-08-19.0 extracts (buildings, roads, places), the census tract polygons, ACS 5-year housing units (B25001) and the strata tables. Nothing is downloaded by hand and nothing outside the challenge package enters the scored file.

What we measure exactly, per tract

Component	Overture side, measured as the score measures it	Evidence for the withdrawn reference
Transport	Length of motorway, trunk, primary and secondary segments cut to the tract polygon, in EPSG:5070	Length of numbered routes (Interstate, US, state, state sub-systems, county) cut to the tract, and again to the tract grown by 30 m to catch routes on the border
Building	Footprints counted by bounding-box centre, the score's rule	USFS Wildfire Risk to Communities building count (strata column usfs_BuildingCount_sum)
Facilities	Places whose categories.primary is fire_department, ambulance_and_ems_services, or one of the six school categories	Population and land area (strata), and the Overture counts themselves
Establishments	All Overture places in the tract	County totals of Overture places, population and large-building counts

How each withdrawn reference is estimated

- **Transport.** TIGER-like length = Interstate + US + state routes + state sub-systems + 0.8 times the route length that runs along the tract border. TIGER border roads lie on the tract edge, so a cut puts them in both neighbouring tracts. In Texas the sub-system weight is 0.45, because farm-to-market roads are only partly TIGER primary or secondary. The gap is the expected value of 1 - min(1, Overture km / TIGER km) under lognormal uncertainty on the TIGER length (sigma 0.35). The chance that the component is defined rises with the route length.
- **Building.** Microsoft count = 1.03 times the USFS building count, lognormal sigma 0.12. The gap is the expected value of 1 - min(1, Overture count / Microsoft count).
- **Facilities.** For each type, the chance that the tract holds a reference facility comes from a Poisson prior on population and land area, updated by Bayes with what Overture shows (a miss rate and a false-positive rate). The gap given presence compares the Overture count with a truncated Poisson count. The facility half is the expected mean over the defined types, with all eight defined or undefined combinations enumerated and fire and EMS correlated.

- **Establishments.** County establishments = kappa times the county's Overture places, spread within the county half by population and half by large buildings (500 m2 and over). The gap is the expected value of 1 - min(1, places / establishments), lognormal sigma 0.6.
- **Composite.** The expectation, over whether the transport component is defined, of the mean of the defined components.

Check against published facts

The organisers published how many tracts have no TIGER highway. Our route-presence rule reproduces those shares without any label: maricopa-az 0.529 against 0.546 published, northern-ca 0.365 against 0.369, eastern-ok 0.222 against 0.212. South-central-tx gives 0.204 against 0.284 when every farm-to-market road is counted, which is why the Texas sub-system weight is below one.

Edge cases

Every edge case follows one rule from the official definition: a missing reference makes a component undefined, while missing Overture data against a present reference makes it a full gap of 1.

Case	How the score treats it	What our estimate does
Zero-population tracts	Population is not in the formula; the tract is scored whenever a reference layer has something in it	Facility priors fall back to land area alone; the establishment half is treated as undefined when no establishments can be allocated; roads and buildings are estimated as usual
Water-dominated tracts (ALAND = 0)	Seven 99xx water tracts in south-central-tx have no scorable reference and are absent from the sample submission	Land-area priors are zero; a tract with no Overture building but a positive USFS count gets an expected building gap near 1
No Overture data in a component	A positive reference with no Overture feature gives a gap of 1	The expectation is driven by the chance that the reference exists, so a tract with zero Overture fire stations carries the full gap times that chance

Case	How the score treats it	What our estimate does
Undefined components	The composite is the mean of the defined components only; the divisor is 1, 2 or 3	The divisor is taken in expectation: 1/3 when TIGER is likely present, 1/2 when not, weighted by that probability
Eastern Washington	The submission must hold all 9,794 tracts, but no eastern-wa tract is on the public board	Scored by the same model, with the mean of the other regions' intercepts
Grader input	The grader reads the second column	Files are written as two columns, GEOID and coverage_gap_score, GEOID as 11-character text
Data-sparse tracts (few or no Overture features)	A present reference with few Overture features gives a large gap	The estimate leans on its priors: population and land area for facilities, the USFS building count for buildings, the county share for establishments. With no numbered route in the tract, the transport component is most likely undefined, so the divisor drops to 2

Calibration and alternative weightings

The expected-value model gets the shape of each component right but not its scale: on its own it predicts a mean gap of 0.128 against a true public mean of 0.060. We fix the scales with a handful of our own leaderboard scores, each used only as one aggregate number.

The algebra

For any uploaded file s, the public RMSE gives one exact moment of the hidden target y over the public rows:

MSE(s) = mean(y²) - 2 mean(y s) + mean(s²), each mean taken over the public tracts.

- An all-zeros file scored 0.107246299, so the mean of y squared is 0.011502.
- A constant 0.10 file scored 0.097675731, so the public mean gap is 0.0598.
- Constant files on one region at a time give the region means: maricopa-az 0.042, northern-ca 0.068, eastern-ok and south-central-tx together 0.063. A file that changed only eastern-wa tracts scored exactly like all zeros, so no eastern-wa tract is on the

public board.

- Each component's expected contribution is uploaded once, at half scale, from the zero baseline. Its score gives the mean of y times that component; the mean of s squared is computed over the tracts that can be public.
- The weights solve one least-squares system, w = G^-1 c, where G holds the components' cross products and c the measured moments. Its expected public RMSE is the square root of (mean of y squared - w'c).

We spent three uploads on component probes, each the component's expected value at half scale:

Probe (submission)	Public RMSE	Mean of y times the component	Implied correlation with y
Transport (zvM7zKQZ)	0.093959899	0.003254	0.53
Fire and EMS (Yi7w4eeY)	0.100916800	0.001518	0.08
Establishments (2XoZFaDy)	0.103158904	0.001975	-0.05

Transport carries almost all the signal, fire and EMS a little, and our establishments estimate none. The building and school probes were skipped: the building gap is near zero everywhere and the school type rarely sets the facilities half.

With the three region intercepts, the solved weights are 1.061 for transport, 0.541 for fire and EMS and -0.064 for establishments, with intercepts of 0.014 (maricopa-az), 0.041 (northern-ca) and 0.030 (eastern-ok and south-central-tx). The expected public RMSE was 0.0750, against 0.0977 for the best constant; uploaded, blend A scored 0.073886708 (6AMWPqPS). Blend B shrinks the weights toward the uncalibrated model (ridge 0.1, expected 0.0755) as a hedge for the private board; it scored 0.074494132 (tDuujvN9). Eastern Washington, absent from the public board, takes the mean of the three intercepts.

Alternative weightings we considered

Weighting	Why we did not use it
Raw expected values, no calibration	Over-predicts twice over (mean 0.128 against 0.060)

Weighting	Why we did not use it
A fixed divisor of 3, undefined components counted as zero	The official rule is the mean of the defined components; a tract with no highway would be scored as better covered than it is
Population-weighted fitting	The metric is a per-tract RMSE; weighting by people would optimise a different quantity
Many probes per region and stratum	Each probe costs one of five daily uploads and every extra weight is fitted on about 2,800 public tracts; we keep the system small so it transfers to the private board

Compliance statement

The scored file uses only the challenge package and our own public leaderboard scores; the withdrawn reference layers were never downloaded, reconstructed or used.

Used	Not used
Overture 2026-08-19.0 extracts in the challenge bucket (buildings, roads, places)	TIGER/Line roads, from the bucket or anywhere else
Census tract polygons, ACS housing units (B25001), strata tables	Microsoft building footprints
Our own public leaderboard scores, as aggregate statistics of the public target	HIFLD or USGS facility layers, County Business Patterns
Facts printed in the challenge text and dataset README (scoring rules, counts of tracts with no highway)	Any coverage-gap file or copy of one, and any other participant's files, predictions or scores

For the Bias Discovery only, the rules allow outside public data if it is documented. We use two: the USDA SNAP retailer data and HUD Colonias Communities, each listed with its URL and retrieval date under Reproducibility. Neither enters the scored file.